

Julieta Ester Pinto García 769024742

Repositorio: https://github.com/julietapinto/compiladores_umg.git

Aplicación en la nube: [Registro de Visitantes](#)

Servicio en la nube

Objetivo:

Hacer una aplicación personalizada con dos capas por lo menos en la nube.

Propuesta:

Proyecto de “Sistema de Registro de Visitantes”

Arquitectura

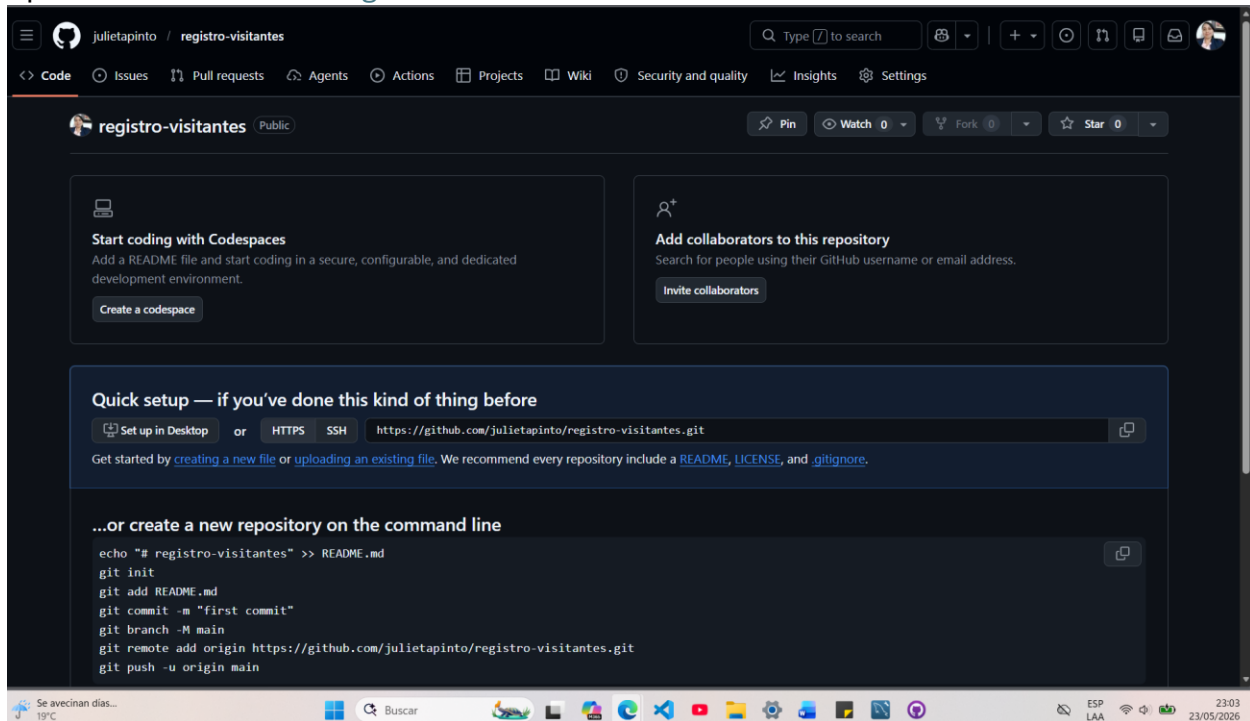
Parte	Herramienta
Frontend	HTML/CSS/JS
Backend	Flask
Base de datos	MySQL
Frontend cloud	Vercel
Backend cloud	Render
DB cloud	Railway

Repositorio

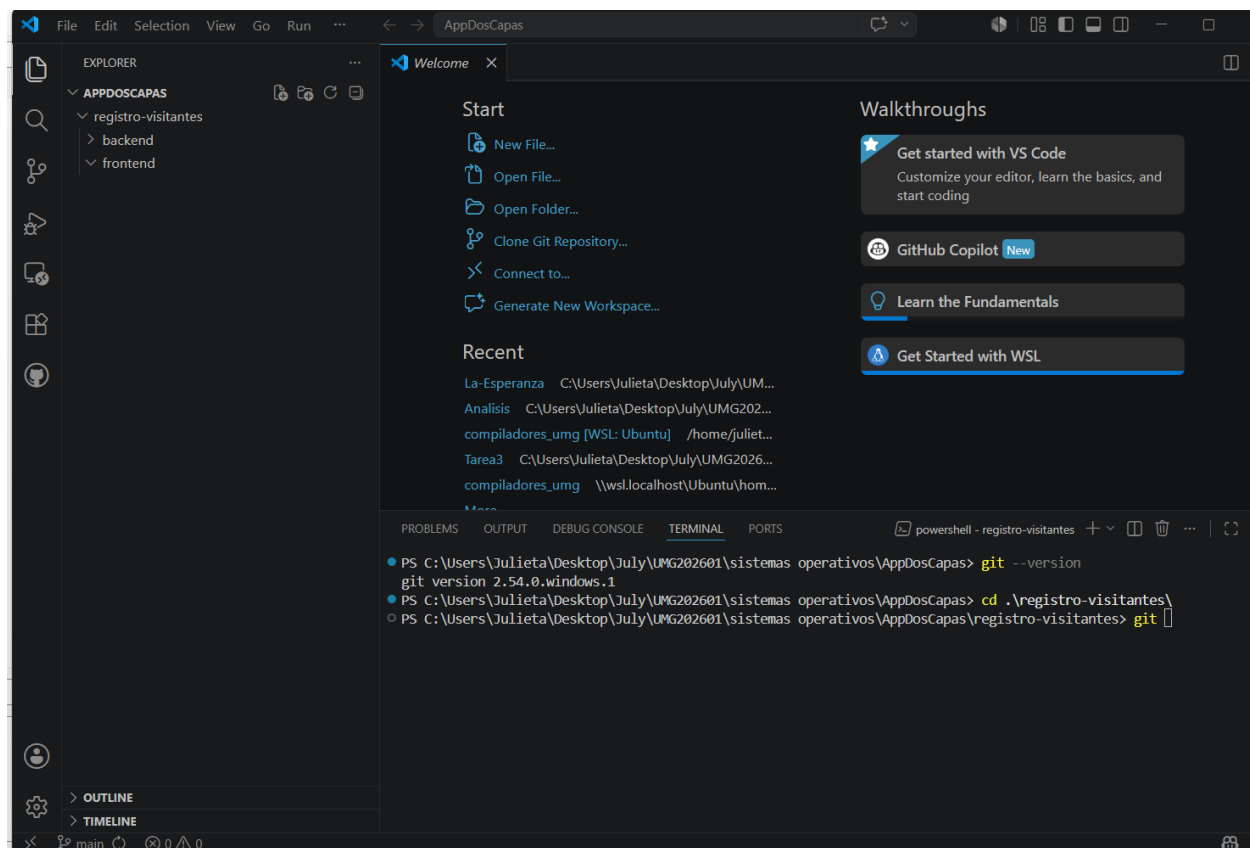
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Clonado

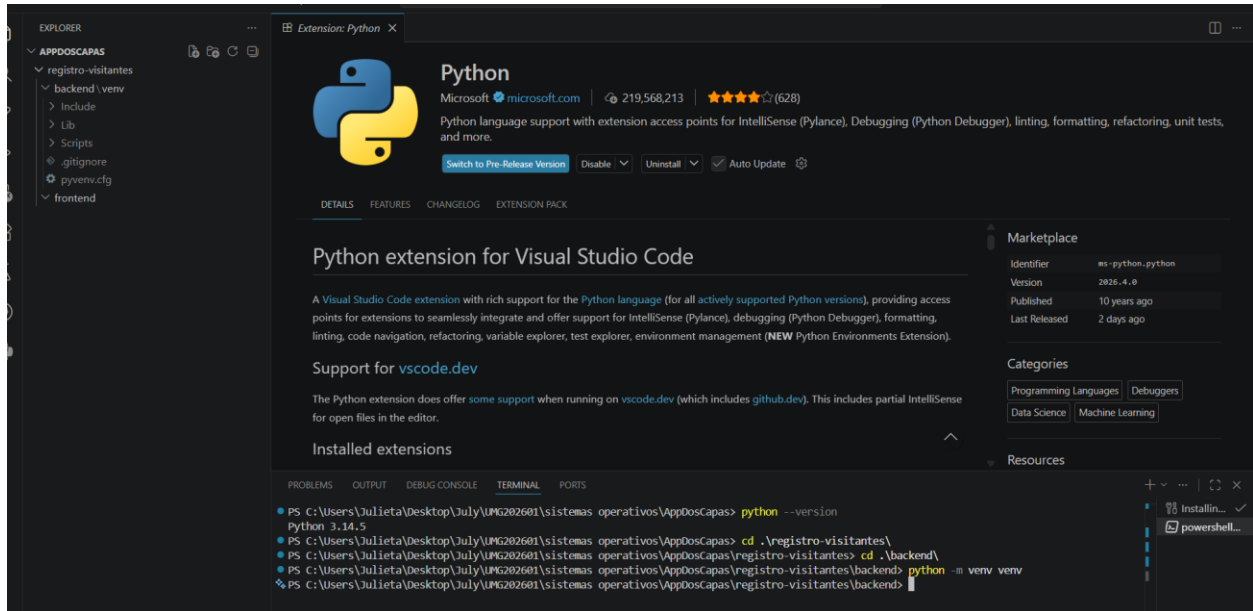


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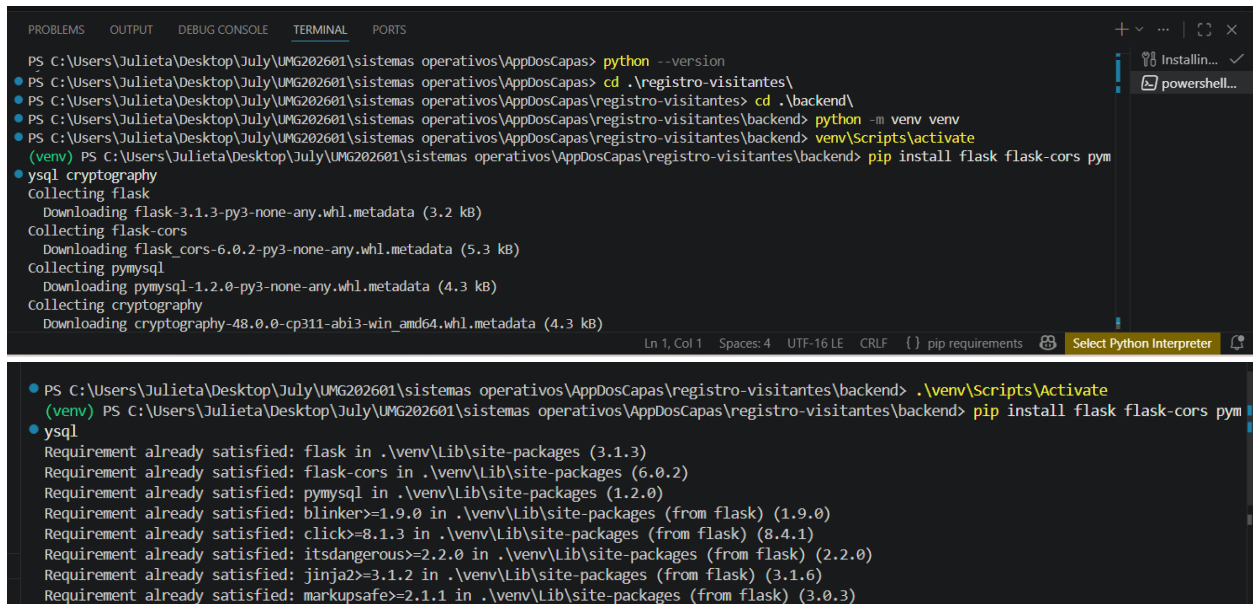
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Creando entorno virtual en backend



Instalando dependencias de flask core

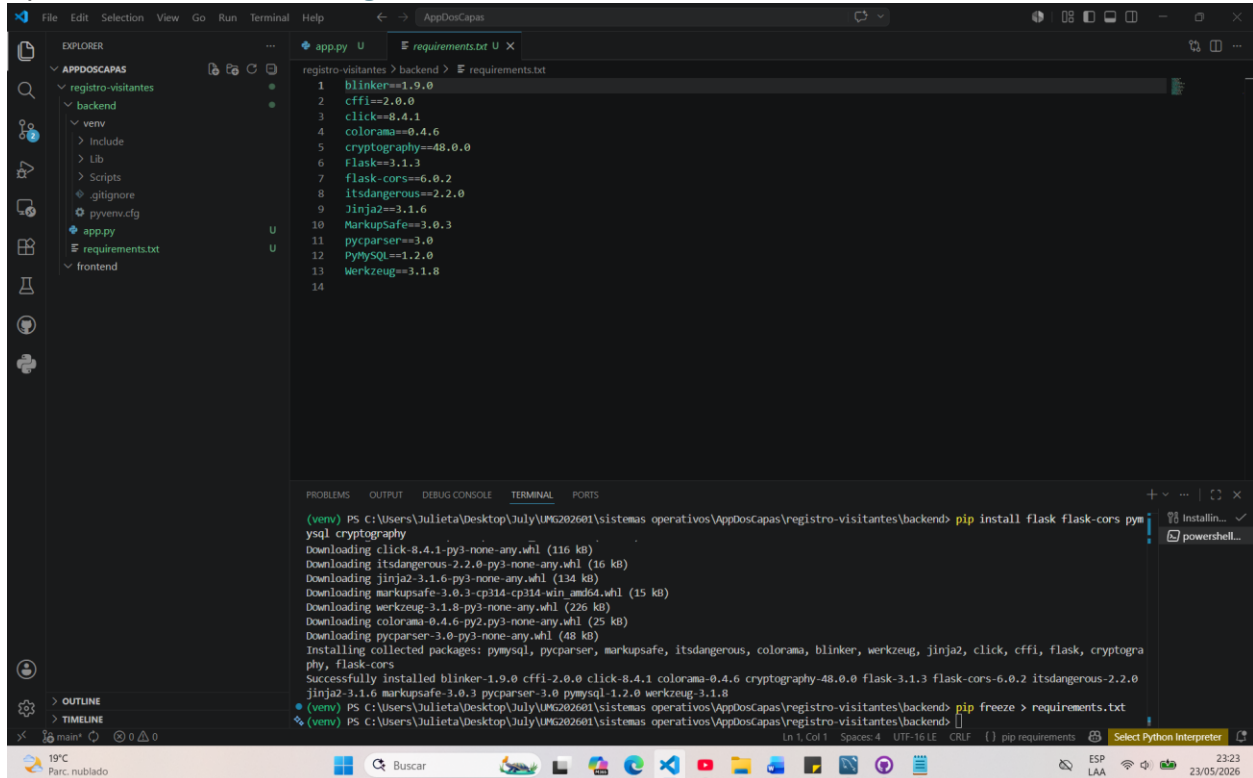


Creación de archivo requirements

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```
File Edit Selection View Go Run Terminal Help
registro-visitantes > backend > requirements.txt
1 blinker==1.9.0
2 cffi==2.0.0
3 click==8.4.1
4 colorama==0.4.6
5 cryptography==48.0.0
6 Flask==3.1.3
7 flask-cors==6.0.2
8 itsdangerous==2.2.0
9 Jinja2==3.1.6
10 MarkupSafe==3.0.3
11 pycparser==3.0
12 PyMySQL==1.2.0
13 Werkzeug==3.1.8
14

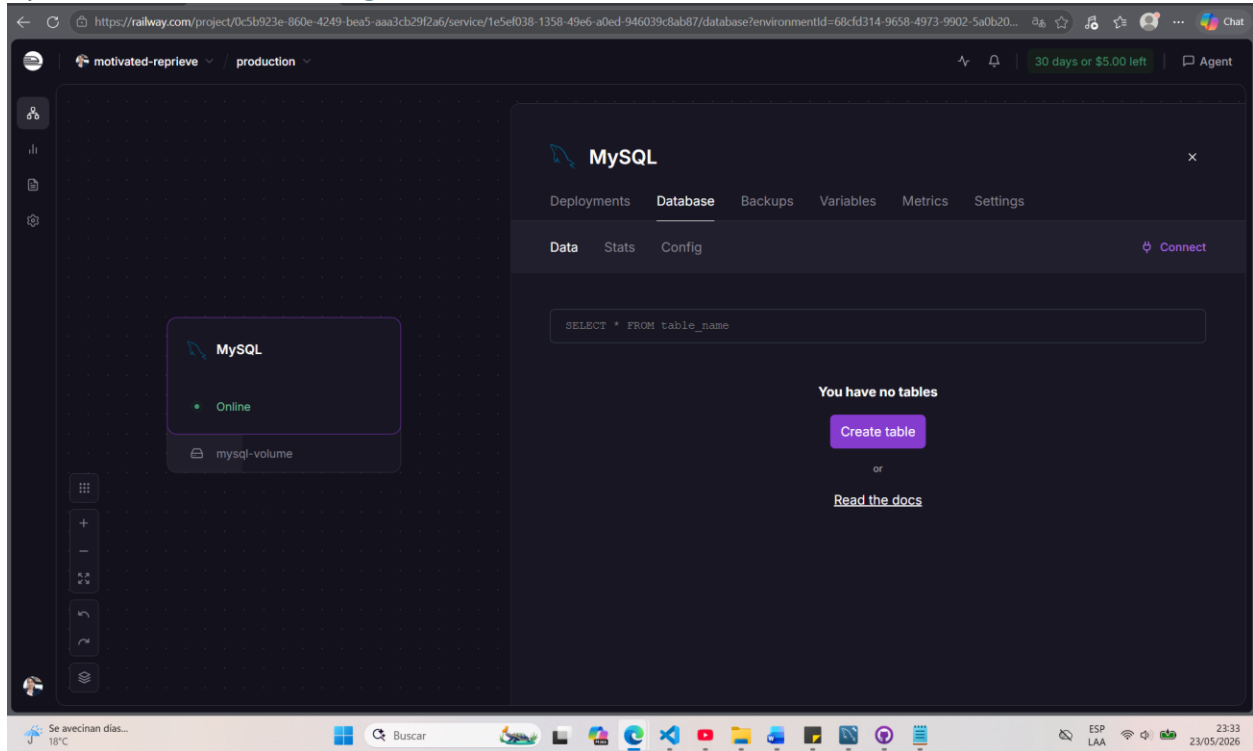
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(venv) PS C:\Users\Julieta\Desktop\July\UMG202601\sistemas operativos\AppDoscapas\registro-visitantes\backend> pip install flask flask-cors pymysql cryptography
Downloading click-8.4.1-py3-none-any.whl (116 kB)
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloading Jinja2-3.1.6-py3-none-any.whl (134 kB)
Downloading MarkupSafe-3.0.3-cp314-cp314-win_amd64.whl (15 kB)
Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading pycparser-3.0-py3-none-any.whl (48 kB)
Installing collected packages: pymysql, pycparser, markupsafe, itsdangerous, colorama, blinker, werkzeug, Jinja2, click, cffi, flask, cryptography, flask-cors
Successfully installed blinker-1.9.0 cffi-2.0.0 click-8.4.1 colorama-0.4.6 cryptography-48.0.0 flask-3.1.3 flask-cors-6.0.2 itsdangerous-2.2.0 Jinja2-3.1.6 markupsafe-3.0.3 pycparser-3.0 pymysql-1.2.0 werkzeug-3.1.8
(venv) PS C:\Users\Julieta\Desktop\July\UMG202601\sistemas operativos\AppDoscapas\registro-visitantes\backend> pip freeze > requirements.txt
(venv) PS C:\Users\Julieta\Desktop\July\UMG202601\sistemas operativos\AppDoscapas\registro-visitantes\backend>
```

Creación de base de datos en Railway (Mysql Cloud)

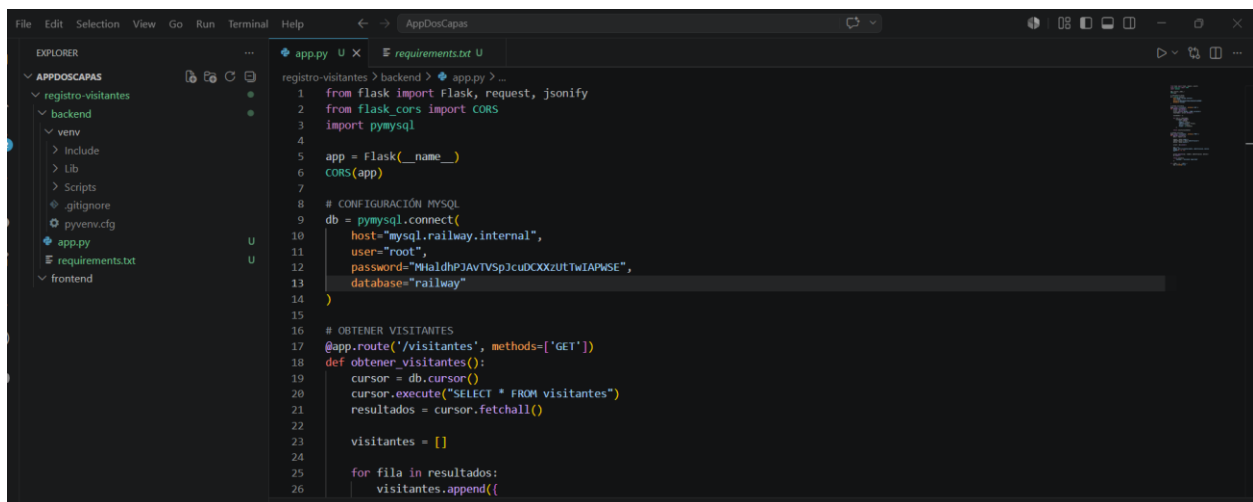
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Configuración de conexión a base de datos en la nube

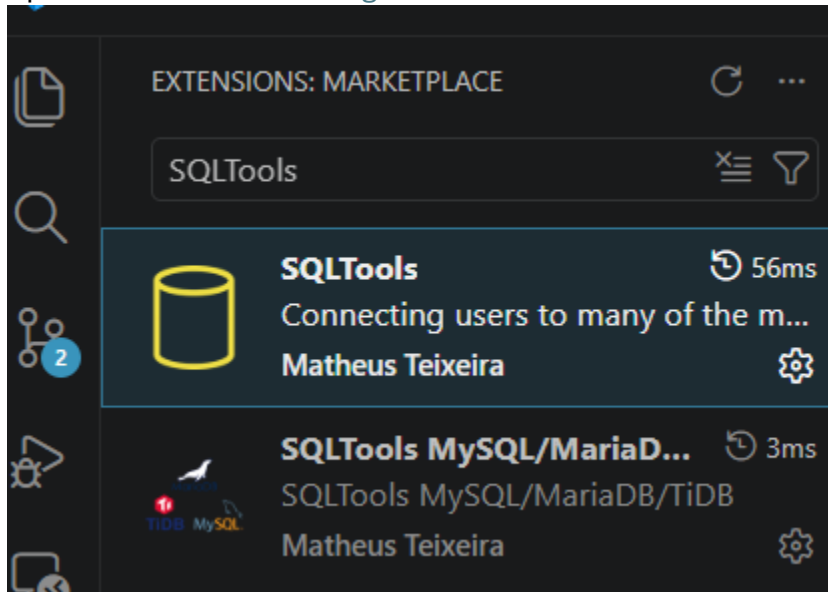


Instalando extensiones necesarias

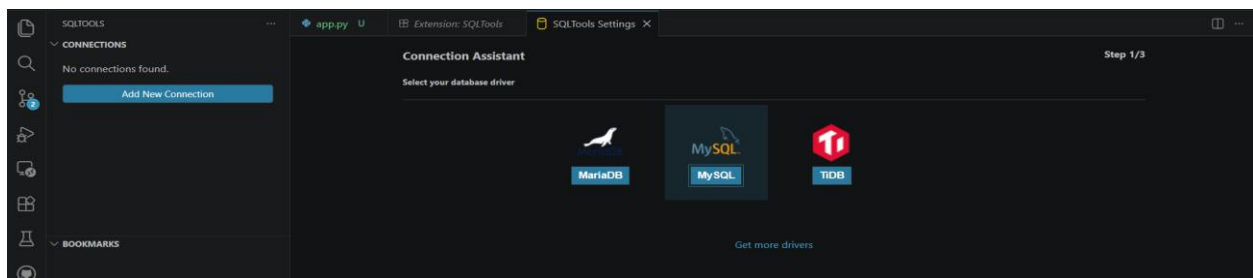
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Creando conexión para sql tolos



Conexión con railway desde visual studio

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The screenshot shows the SQLTools Settings dialog box for a MySQL connection. The dialog is titled "SQLTools Settings" and has a close button (X). It contains several configuration fields:

- Connection group:** (empty text field)
- Connect using*:** (dropdown menu showing "Server and Port")
- Server Address*:** (text field with "kodama.proxy.rlwy.net")
- Port*:** (text field with "34642")
- Database*:** (text field with "railway")
- Username*:** (text field with "root")
- Password mode:** (dropdown menu showing "Save as plaintext in settings")
- Password*:** (password field with masked characters)

Below these fields is a section titled "MySQL driver specific options" which contains:

- Authentication Protocol:** (dropdown menu showing "default") with a note: "Try to switch protocols in case you have problems."
- SSL:** (dropdown menu showing "Disabled")

At the bottom of the settings section are:

- Over SSH:** (dropdown menu showing "Disabled")
- Connection Timeout:** (empty text field)
- Show records default limit:** (text field with "100")

A green status bar at the bottom of the settings section says "Successfully connected!". Below that is a blue button labeled "SAVE CONNECTION".

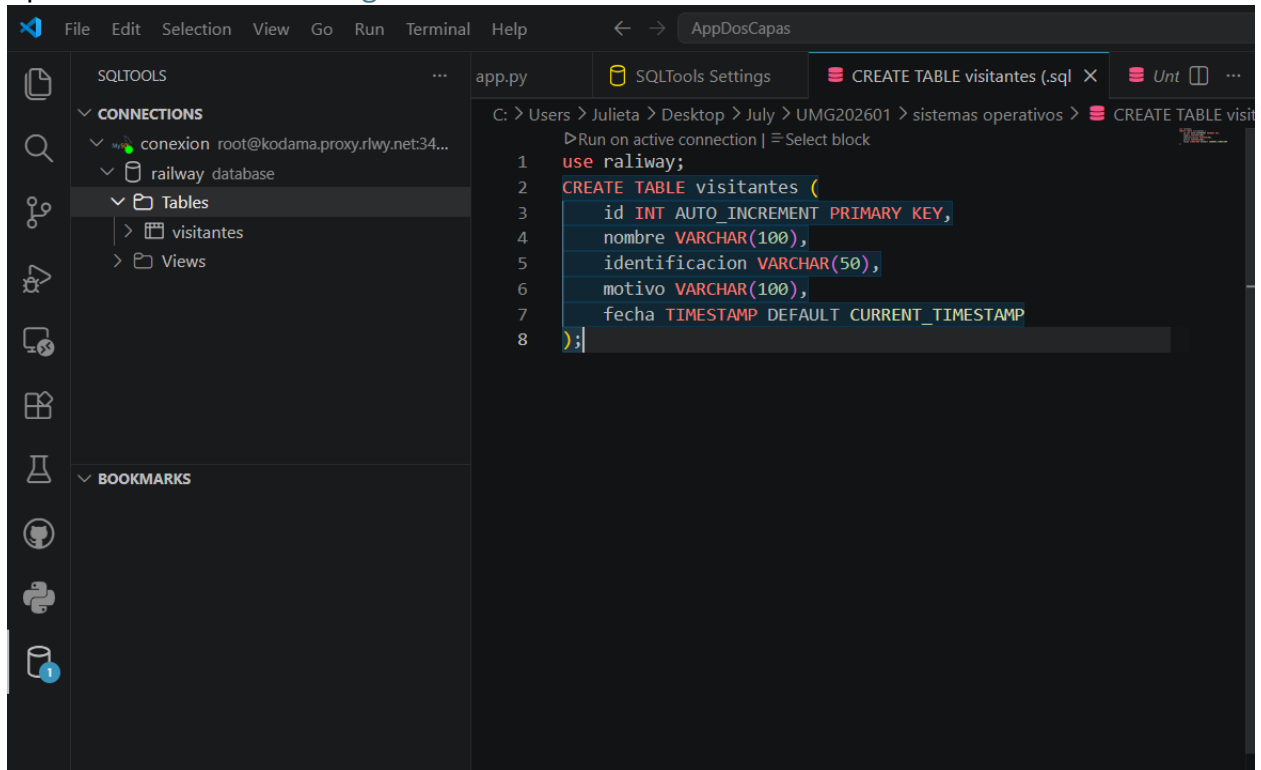
The bottom of the screenshot shows the "OUTPUT" tab of the IDE, displaying an error message: "[1779601434281] ERROR (ext): ERROR: Error selecting connection SQLTools Driver Credentials: Password".

Tabla creada

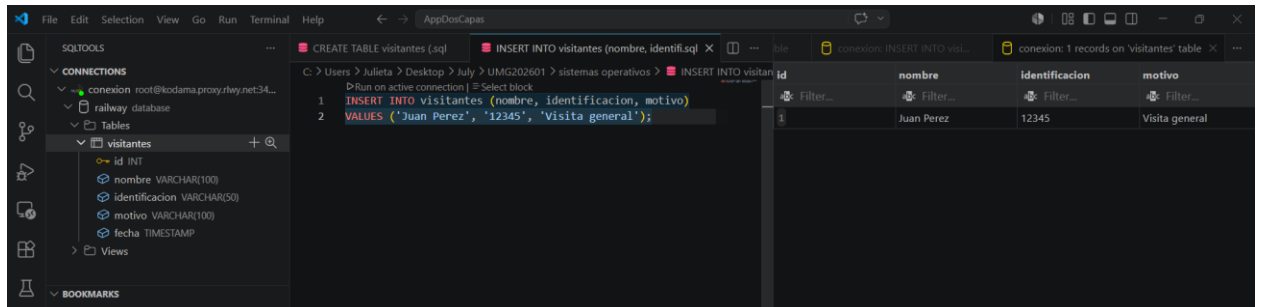
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Dato insertado

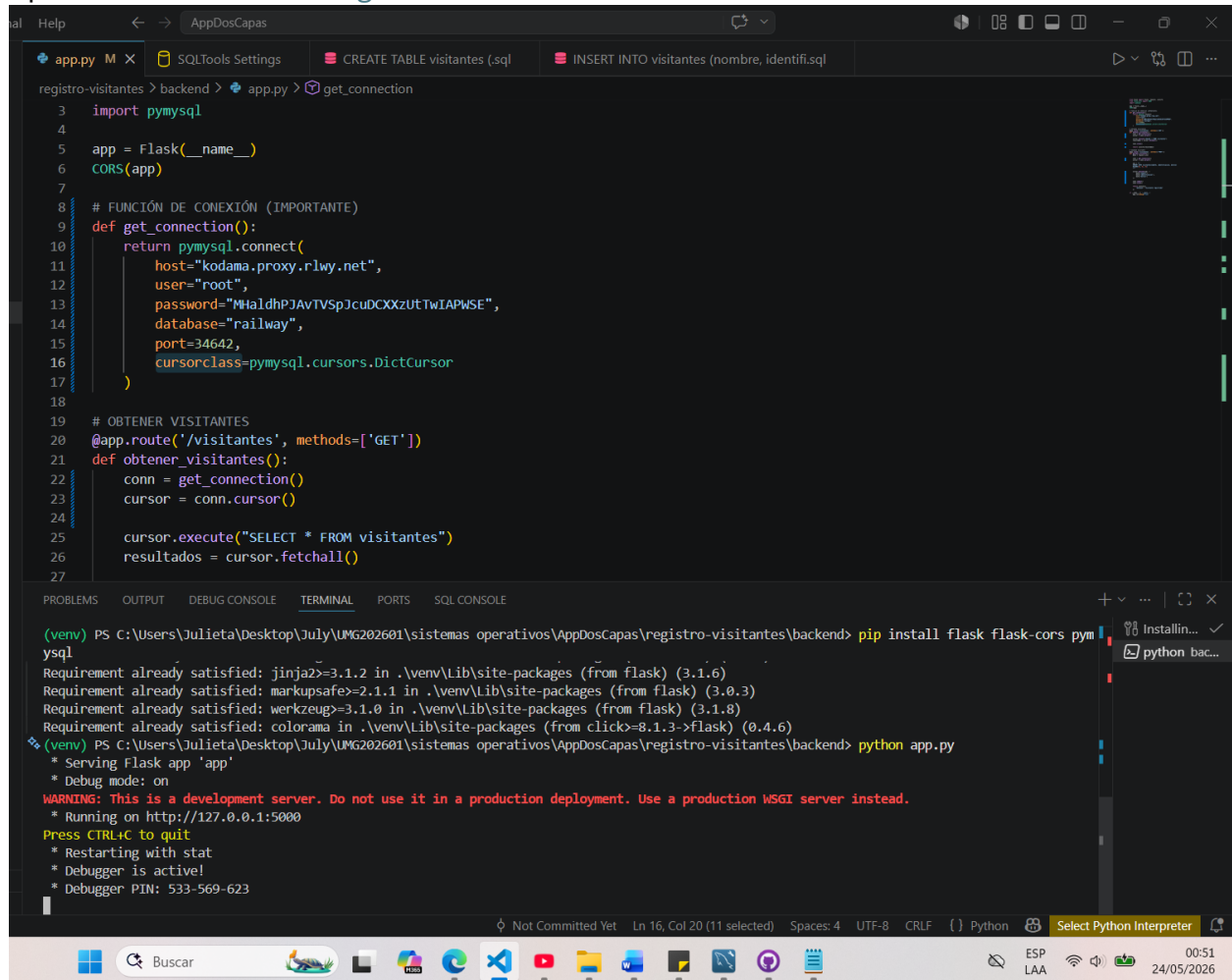


Ejecutando aplicación

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The screenshot shows a Visual Studio Code editor with a Python file named `app.py` open. The code is a Flask application that connects to a MySQL database and serves a GET endpoint for `/visitantes`. The terminal at the bottom shows the command `pip install flask flask-cors pymysql` being executed, followed by `python app.py`, which starts the Flask development server on `http://127.0.0.1:5000`. The status bar at the bottom indicates the file is not committed, the cursor is at line 16, column 20, and the Python interpreter is selected.

```
app.py M x SQLTools Settings CREATE TABLE visitantes (.sql) INSERT INTO visitantes (nombre, identifi.sql
registro-visitantes > backend > app.py > get_connection
3 import pymysql
4
5 app = Flask(__name__)
6 CORS(app)
7
8 # FUNCIÓN DE CONEXIÓN (IMPORTANTE)
9 def get_connection():
10     return pymysql.connect(
11         host="kodama.proxy.rlwy.net",
12         user="root",
13         password="MhaldhPJAvTVSpJcuDCXXzUTwIAPWSE",
14         database="railway",
15         port=34642,
16         cursorclass=pymysql.cursors.DictCursor
17     )
18
19 # OBTENER VISITANTES
20 @app.route('/visitantes', methods=['GET'])
21 def obtener_visitantes():
22     conn = get_connection()
23     cursor = conn.cursor()
24
25     cursor.execute("SELECT * FROM visitantes")
26     resultados = cursor.fetchall()
27
```

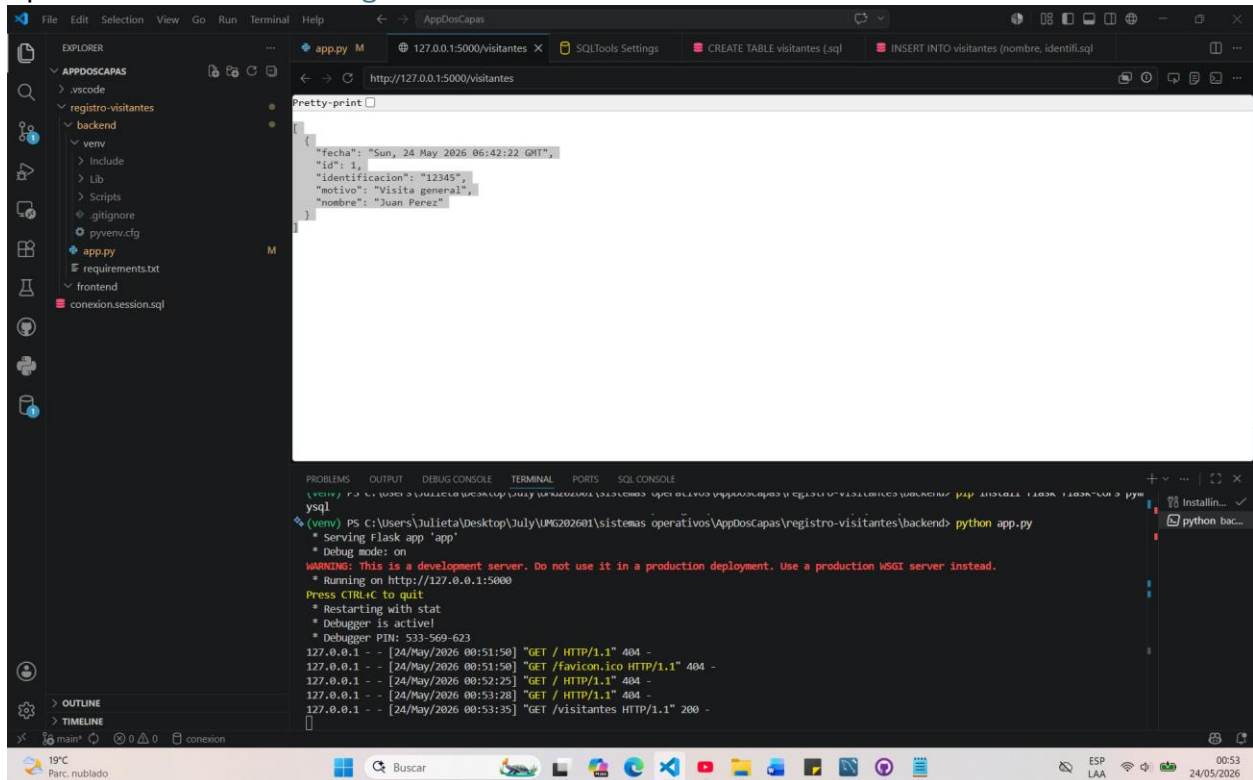
```
(venv) PS C:\Users\Julieta\Desktop\July\UMG202601\sistemas operativos\AppDosCapas\registro-visitantes\backend> pip install flask flask-cors pymysql
Requirement already satisfied: Jinja2>=3.1.2 in .\venv\Lib\site-packages (from flask) (3.1.6)
Requirement already satisfied: MarkupSafe>=2.1.1 in .\venv\Lib\site-packages (from flask) (3.0.3)
Requirement already satisfied: Werkzeug>=3.1.0 in .\venv\Lib\site-packages (from flask) (3.1.8)
Requirement already satisfied: Colorama in .\venv\Lib\site-packages (from click>=8.1.3->flask) (0.4.6)
(venv) PS C:\Users\Julieta\Desktop\July\UMG202601\sistemas operativos\AppDosCapas\registro-visitantes\backend> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 533-569-623
```

Probando petición `http://127.0.0.1:5000/visitantes`

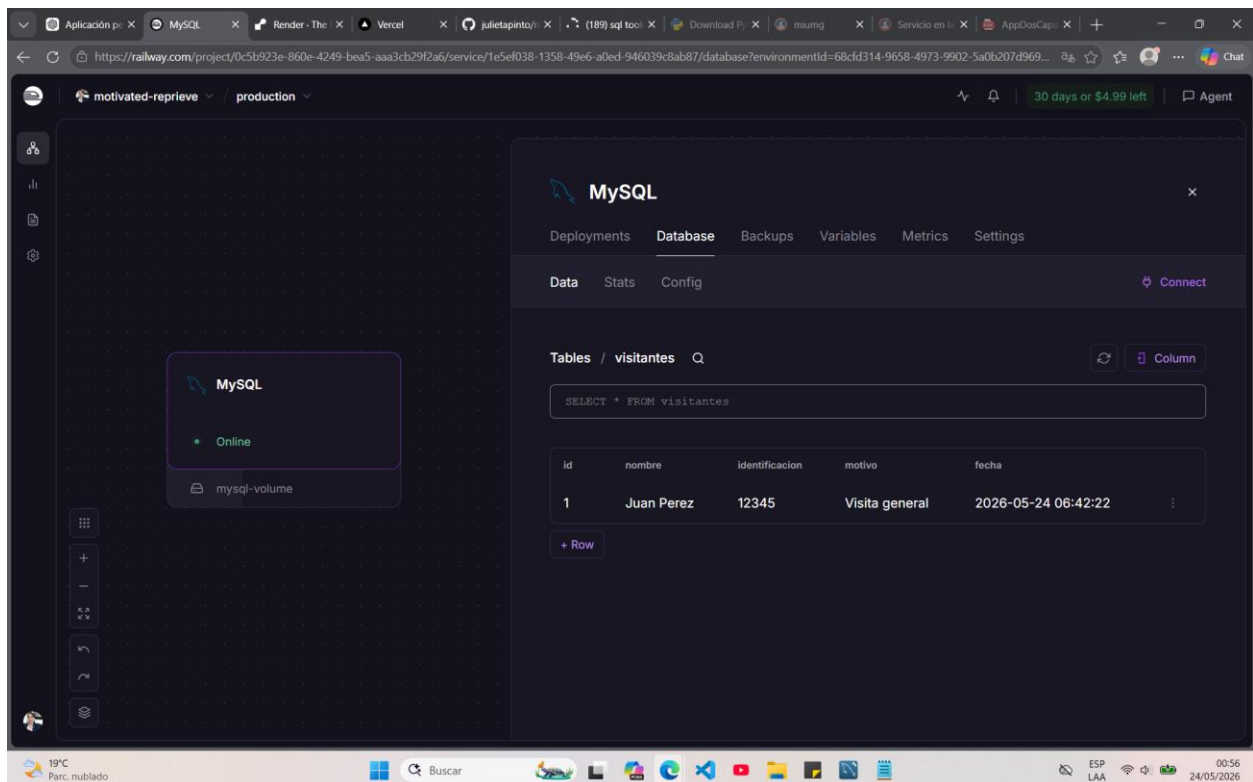
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Bse de datos en la nube funcionando

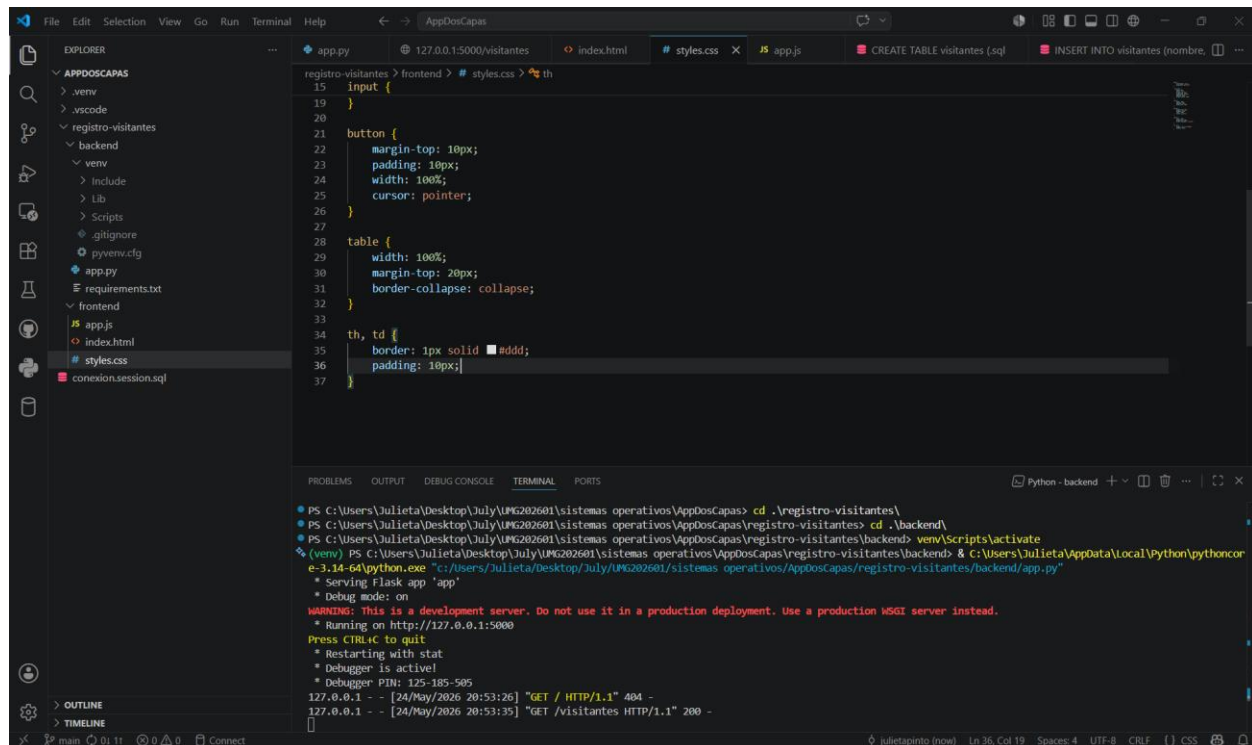


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Capa de estilos

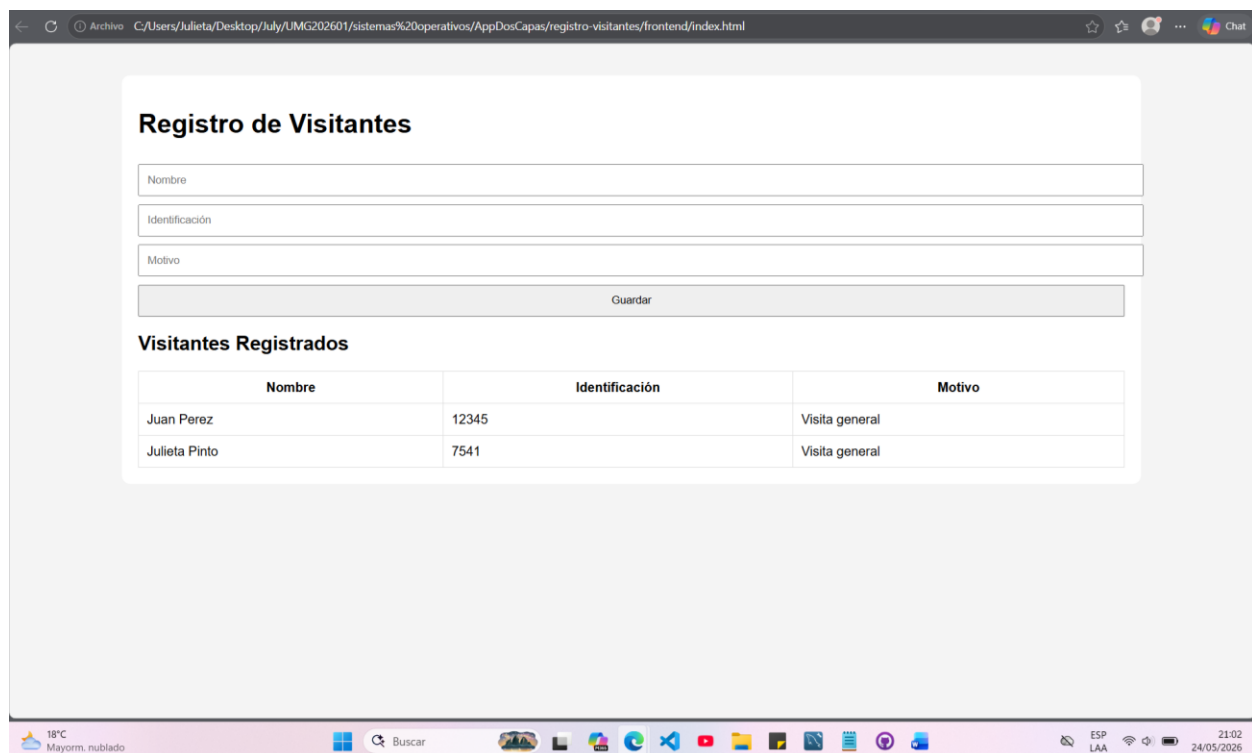


The screenshot shows the Visual Studio Code editor with the 'styles.css' file open. The file contains CSS rules for an input field, a button, and a table. The terminal window at the bottom shows the command prompt and the output of the Flask application, including the URL 'http://127.0.0.1:5000' and the status 'Running on http://127.0.0.1:5000'.

```
registro-visitantes > frontend > # styles.css > th
15 input {
16   width: 100%;
17   height: 30px;
18 }
19
20
21 button {
22   margin-top: 10px;
23   padding: 10px;
24   width: 100%;
25   cursor: pointer;
26 }
27
28 table {
29   width: 100%;
30   margin-top: 20px;
31   border-collapse: collapse;
32 }
33
34 th, td {
35   border: 1px solid #ddd;
36   padding: 10px;
37 }
```

```
PS C:\Users\Julieta\Desktop\July\UMG202601\sistemas operativos\AppDosCapas> cd .\registro-visitantes\
PS C:\Users\Julieta\Desktop\July\UMG202601\sistemas operativos\AppDosCapas\registro-visitantes> cd .\backend\
PS C:\Users\Julieta\Desktop\July\UMG202601\sistemas operativos\AppDosCapas\registro-visitantes\backend> venv\Scripts\activate
(venv) PS C:\Users\Julieta\Desktop\July\UMG202601\sistemas operativos\AppDosCapas\registro-visitantes\backend> & C:\Users\Julieta\AppData\Local\Python\pythoncon
e-3.14-64\python.exe "c:/Users/Julieta/Desktop/July/UMG202601/sistemas operativos/AppDosCapas/registro-visitantes/backend/app.py"
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 125-185-505
127.0.0.1 - - [24/May/2026 20:53:26] "GET / HTTP/1.1" 404 -
127.0.0.1 - - [24/May/2026 20:53:35] "GET /visitantes HTTP/1.1" 200 -
```

Probando aplicación

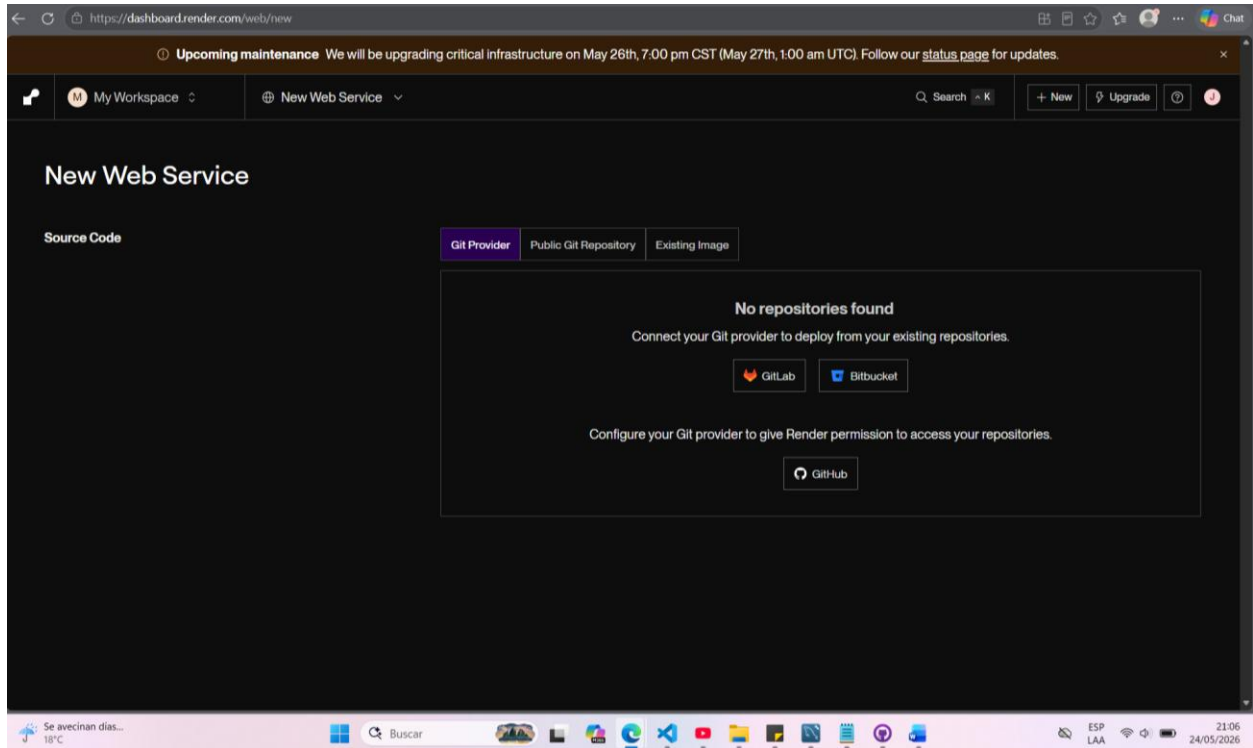


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Configurando en Render para web service



Configuración en Render

My Workspace | New Web Service | Search | + New | Upgrade | Chat

Name
A unique name for your web service.
registro-visitantes

Language
Choose the runtime environment for this service.
Python 3

Branch
The Git branch to build and deploy.
main

Region
Your services in the same region can communicate over a private network.
Ohio (US East)

Root Directory Optional
If set, Render runs commands from this directory instead of the repository root. Additionally, code changes outside of this directory do not trigger an auto-deploy. Most commonly used with a monorepo.
backend

Build Command
Render runs this command to build your app before each deploy.
backend/ \$ pip install -r requirements.txt

Start Command
Render runs this command to start your app with each deploy.
backend/ \$ gunicorn app:app

Instance Type

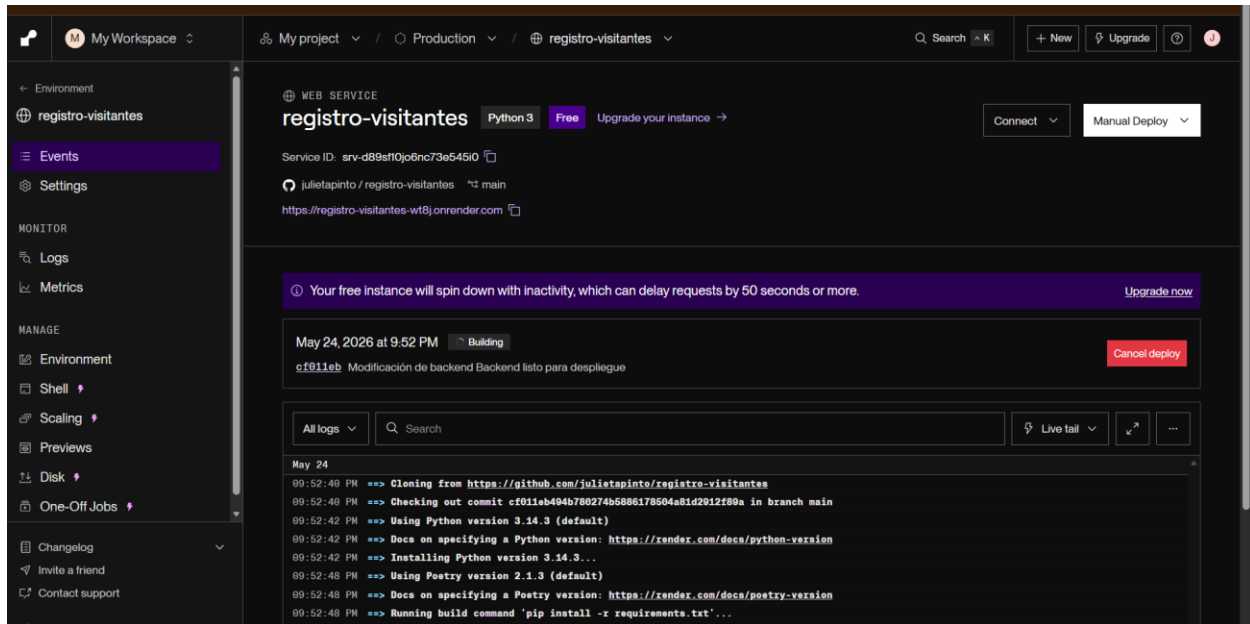
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Deployed en Render Exitoso

El link de Render sirve para que backend Flask esté en internet. Ahora funciona desde cualquier lugar con una URL pública. El frontend usará ese link para obtener datos de la API (como /visitantes)

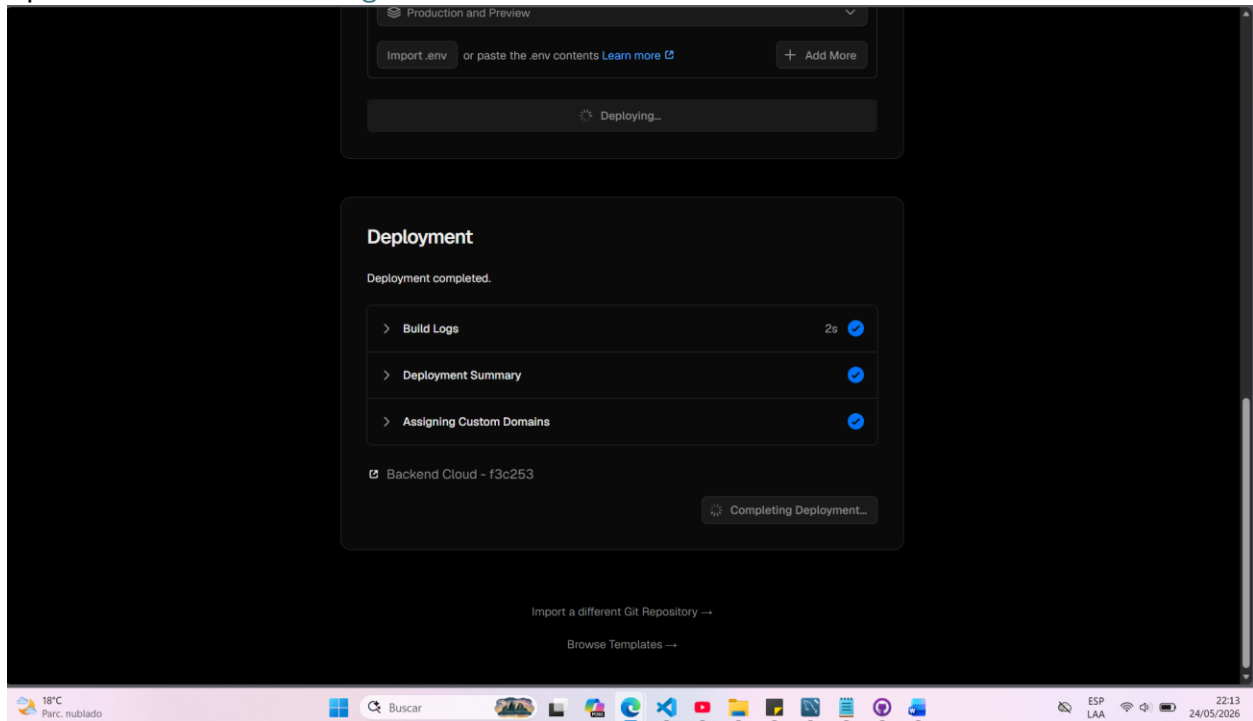


Desplegando app en vercel

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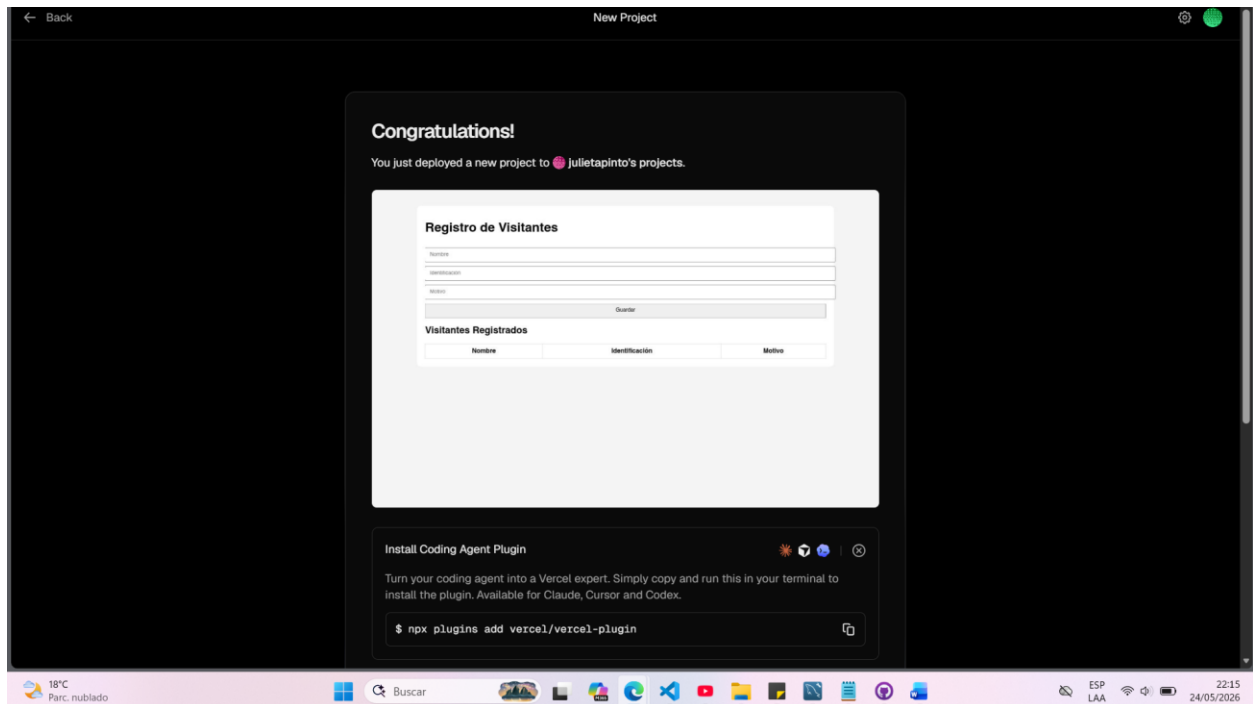
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App montada en internet

[Registro de Visitantes](#)



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Aplicación en la nube: [Registro de Visitantes](https://registro-visitantes.vercel.app)

https://registro-visitantes.vercel.app

Registro de Visitantes

Nombre

Identificación

Motivo

Guardar

Visitantes Registrados

Nombre	Identificación	Motivo
Juan Perez	12345	Visita general
Julieta Pinto	7541	Visita general
Ana Lucia	789	Visita externa
Edson	Gil	Visita Extemporanea

18°C

79°F